

Title

Prenatal Aerobic Exercise Reduces Postpartum Depression: Findings From a Randomized Controlled Trial of Pregnant Women

Abstract

Physical activity during pregnancy may confer psychological benefits that extend into the postpartum period. We randomized 398 pregnant women with uncomplicated pregnancies to either structured aerobic exercise (three 60-minute sessions weekly through mid-gestation) or routine prenatal care. At three months postpartum, women in the exercise group exhibited substantially lower depressive symptom scores on validated instruments compared to the standard-care group. The proportion of women meeting screening criteria for possible postpartum depression was significantly reduced in the exercise group. Clinical diagnoses of postpartum depression were correspondingly lower in the active group. Maternal and neonatal outcomes including gestational complications, delivery method, and neonatal health indicators were similar between groups, indicating the exercise intervention was safe. These findings support incorporating aerobic physical activity into prenatal care as a practical, accessible strategy to reduce postpartum depressive symptoms and promote maternal mental health in low-risk pregnancies.